

Material: ALUMINUM 6061-T6 ($S_y = 40,000 \text{ psi}$, $E = 10 \times 10^6 \text{ psi}$)

Applied Load (F): 600 lbf

Factor of Safety (N): 4

Stress analyses

Feature A:

Knowns: SF

$F = 600 \text{ lbf}$

$E = 10 \times 10^6 \text{ psi}$

$SF = 4$

$S_y = 40,000 \text{ psi}$

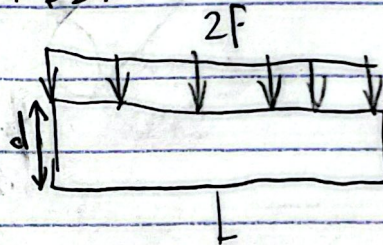
Unknowns:

d

Assumptions:

$L = 1 \text{ in}$

FBD:



Algebraic Solution:

$$\sigma_{\text{allow}} = \frac{S_y}{SF} \rightarrow \frac{S_y}{SF} = \frac{2FL}{2Z} \Rightarrow Z = \frac{2FLSF}{2S_y}$$

$$Z = \frac{\pi d^3}{32} = \frac{2FLSF}{2S_y} \Rightarrow d = \sqrt[3]{\frac{32FLSF}{2\pi S_y}}$$

Numerical Solution:

$$d = \sqrt[3]{\frac{32(600 \text{ lbf})(1 \text{ in})(4)}{2\pi(40,000 \text{ psi})}} \Rightarrow \boxed{d = 0.849 \text{ in}}$$

Feature B:

Knowns:

$$F = 600 \text{ lbf}$$

$$S_y = 40,000 \text{ psi}$$

$$SF = 4$$

Unknowns:

$$d$$

FBD:



Assumptions:

$$L = 0.844 \text{ in}$$

$$h = 1 \text{ in}$$

Algebraic solution:

$$\frac{S_y}{SF} = \frac{2F}{A} \rightarrow \frac{S_y}{SF} = \frac{2F}{d \cdot L} \rightarrow d = \frac{SF \cdot 2 \cdot F}{S_y \cdot L}$$

Numerical solution:

$$d = \frac{(4)(2)(600 \text{ lbf})}{(40,000 \text{ psi})(0.844)} \rightarrow \boxed{d = 0.141 \text{ in}}$$

Feature C:



Knowns:

$F = 600 \text{ lbf}$

$S_y = 40,000 \text{ psi}$

$SP = 4$



Unknowns:

h



Assumptions:

$L = 1 \text{ in}$

$w \approx 3L$

$w = 3h$



Feature D:



Knowns:

$F = 600 \text{ lbf}$

$S_y = 40,000 \text{ psi}$

$SP = 4$



Unknowns:

w

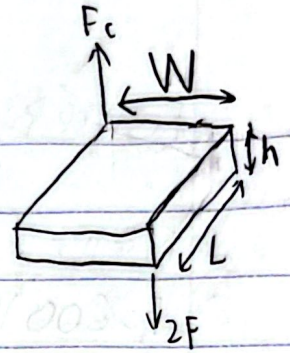
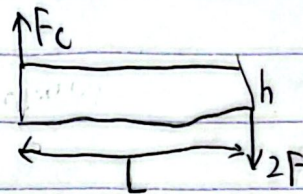


Assumptions:

$L = 1 \text{ in}$

$h = 2 \text{ in}$

FBD:



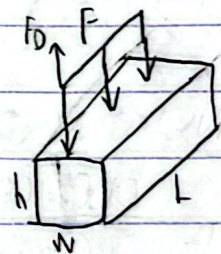
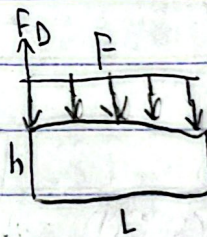
Algebraic solution:

$$\frac{S_y}{SP} = \frac{2FL}{Z} \rightarrow Z = \frac{2FL SP}{S_y} \rightarrow \frac{wh^2}{6} = \frac{2FL SP}{S_y}$$

$$h = \sqrt{\frac{6(2F)L(SP)}{S_y(w)}}$$

Numerical solution:

$$h = \sqrt{\frac{6(1200 \text{ lbf})(1 \text{ in})(4)}{40,000 \text{ psi}(3 \text{ in})}} = \boxed{h = 0.500}$$



FBD:

Algebraic solution:

$$\frac{S_y}{SP} = \frac{FL}{Z} \rightarrow Z = \frac{SP FL}{S_y} \rightarrow \frac{wh^2}{6} = \frac{SP FL}{S_y}$$

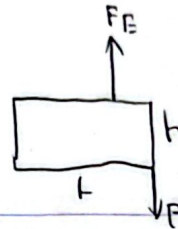
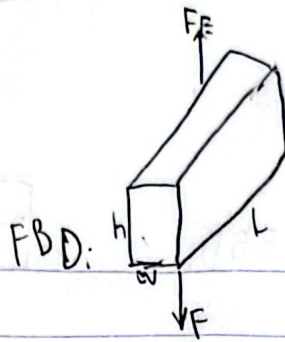
$$w = \frac{6SPFL}{S_y h^2}$$

Numerical solution:

$$w = \frac{6(4)(1 \text{ in})(600 \text{ lbf})}{2(40,000 \text{ psi})(2 \text{ in})^2}$$

$$\boxed{w = 0.045 \text{ in}}$$

Feature E:



FBD:

Knowns:

Algebraic solution:

$$F = 600 \text{ lbf}$$

$$S_y = 70,000 \text{ psi}$$

$$SF = 4$$

$$\frac{S_y}{SF} = \frac{WL}{2} \rightarrow 2 = \frac{Lh^2}{6} \rightarrow h = \sqrt{\frac{6SFwF}{LS_y}}$$

Unknowns:

Numerical solution:

h

$$h = \frac{6(4)(0.045 \text{ in})(600 \text{ lbf})}{1(70,000 \text{ psi})} \rightarrow h = 0.127 \text{ in}$$

Assumptions:

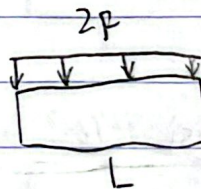
$$L = 1 \text{ in}$$

$$w = 0.045 \text{ in}$$

Stress Analysis:

Feature A:

FBD:



Knowns:

$$\delta_{\max} = 0.005 \text{ in}$$

$$E = 10 \times 10^6 \text{ psi}$$

$$F = 600 \text{ lbf}$$

$$SF = 4$$

Algebraic solution:

$$\frac{\delta}{SF} = \frac{WL^3}{8EI} \rightarrow I = \frac{\pi d^4}{64}$$

$$d = \sqrt[4]{\frac{64SF L^3 2F}{8EI 8\pi}}$$

Unknowns:

Numerical solution:

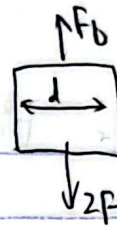
Assumptions:

$$L = 1 \text{ in}$$

$$d = \sqrt[4]{\frac{64(4)(600 \text{ lbf})(1 \text{ in})^3(600 \text{ lbf})}{8(10 \times 10^6 \text{ psi})(0.005 \text{ in})(8\pi)}} \rightarrow d = 2.103 \text{ in}$$

Feature B:

FBD:



Knowns:

Algebraic Solution:

$$E = 10 \times 10^6 \text{ psi}$$

$$\delta_{\max} = \frac{PL}{AE}$$

$$SF = 4$$

$$F = 600 \text{ lbf}$$

$$\frac{\delta}{SF} = \frac{2Fh}{d^2 E}$$
$$d = \sqrt{\frac{2FhSF}{\delta E}}$$

Unknowns:

Numerical Solution:

$$d$$

$$1 \text{ in } (4) (1200 \text{ lbf})$$

Assumptions:

$$d = 0.005 \text{ in } (0.703 \text{ in}) (10 \times 10^6 \text{ psi})$$

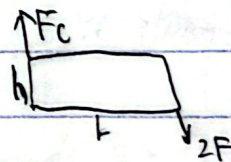
$$L = 0.703 \text{ in}$$

$$d = 0.137 \text{ in}$$

$$h = 1 \text{ in}$$

Feature C:

FBD:



Knowns:

Algebraic Solution:

$$E = 10 \times 10^6 \text{ psi}$$

$$SF = 4$$

$$F = 600 \text{ lbf}$$

$$\frac{\delta}{SF} = \frac{WL^3}{3EI} \rightarrow I = \frac{bh^3}{12} \rightarrow h = \sqrt[3]{\frac{12 \cdot SF \cdot L^3 \cdot 2F}{W \cdot E \cdot \delta}}$$

Unknowns:

Numerical Solution:

$$h$$

$$h = \sqrt[3]{\frac{12(4)(1 \text{ in})^3(1200 \text{ lbf})}{3 \text{ in}(10 \times 10^6 \text{ psi})(0.005 \text{ in})(3)}}$$

Assumptions:

$$L = 1 \text{ in}$$

$$W = 3 \text{ in}$$

$$h = 0.504 \text{ in}$$

Feature D:

Knowns:

$$E = 10 \times 10^6 \text{ psi}$$

$$SF = 4$$

$$F = 600 \text{ lbf}$$

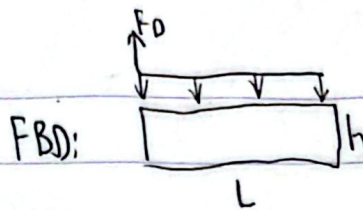
Unknowns:

w

Assumptions:

$$L = 1 \text{ in}$$

$$H = 2 \text{ in}$$



Algebraic solution:

$$\frac{\delta}{SF} = \frac{wL^3}{8EI} \rightarrow I = \frac{wL^3}{12} \rightarrow w = \frac{FL^3 12}{8EH^2}$$

Numerical solution:

$$w = \frac{(600 \text{ lbf})(1 \text{ in})^3 (12)}{8(10 \times 10^6 \text{ psi})(2 \text{ in})^2}$$

$$w = 0.0045 \text{ in}$$

Feature E:

Knowns:

$$E = 10 \times 10^6 \text{ psi}$$

$$SF = 4$$

$$F = 600 \text{ lbf}$$

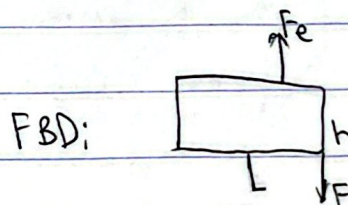
Unknowns:

h

Assumptions:

$$L = 1 \text{ in}$$

$$w = 0.0045 \text{ in}$$



Algebraic solution:

$$\frac{\delta}{SF} = \frac{wL^3}{3EI} \Rightarrow I = \frac{Lh^3}{12} \rightarrow h = \sqrt[3]{\frac{FW^3 12}{3E 8L}}$$

Numerical solution:

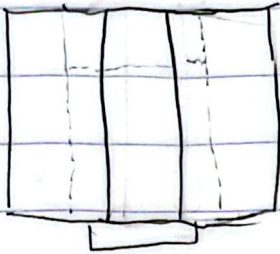
$$h = \sqrt[3]{\frac{(600 \text{ lbf})(0.0045 \text{ in})^3 (12)}{3(10 \times 10^6 \text{ psi})(0.005 \text{ in})(1 \text{ in})}}$$

$$h = 0.0016 \text{ in}$$

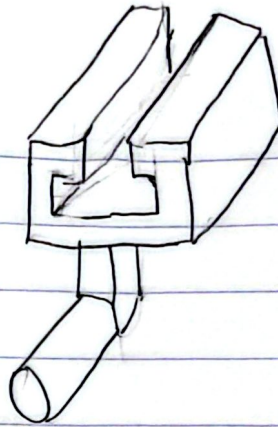
Multiview:

Sketch:

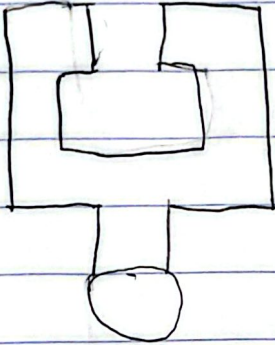
Top



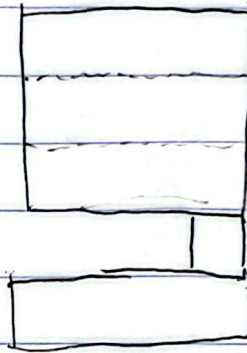
Isometric



Bottom

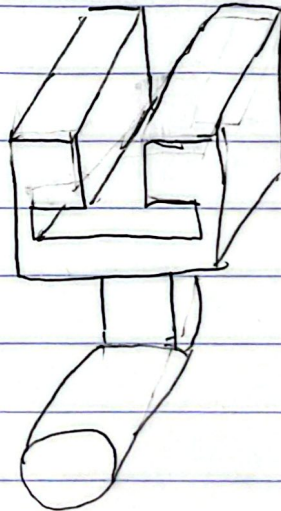
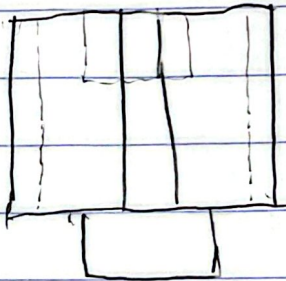


Side

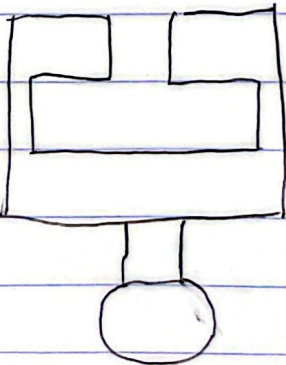


Stiffness:

Top



Bottom



Side

